

RockSurv Petrography

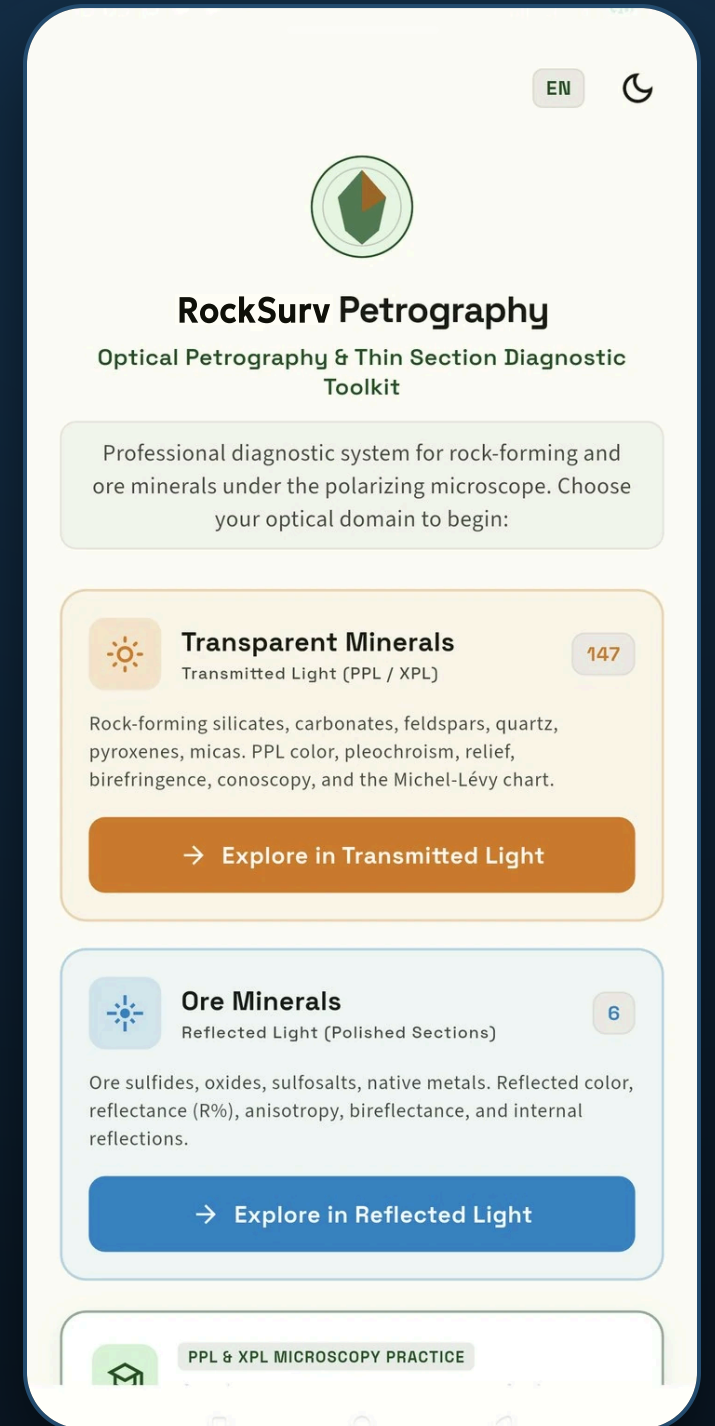
High-Precision Optical Petrography
& Thin Section Mineralogy Suite.

• 147 Transmitted Minerals

• 6 Ore Sulfides & Oxides

• 267+ Reference Micrographs

• 100% Offline Engine



Professional standalone toolkit for polarized light microscopy, thin section identification, and ore petrography.

Dual-Light Framework: Transmitted & Reflected Microscopy

Seamless switching between rock-forming silicates in transmitted light (PPL/XPL) and opaque ore minerals in reflected light (RLM/XRL).

Transmitted Light (147 Minerals)

Complete suite of silicates, carbonates, and sulfates with full crystallographic and optical constants under plane- and cross-polarized light.

Reflected Light (Ore Mineralogy)

Diagnostics for opaque sulfides and oxides evaluated by reflectance percentage (R%), bireflectance, anisotropy, and polishing hardness.

One-Tap Synchronized Mode Switcher

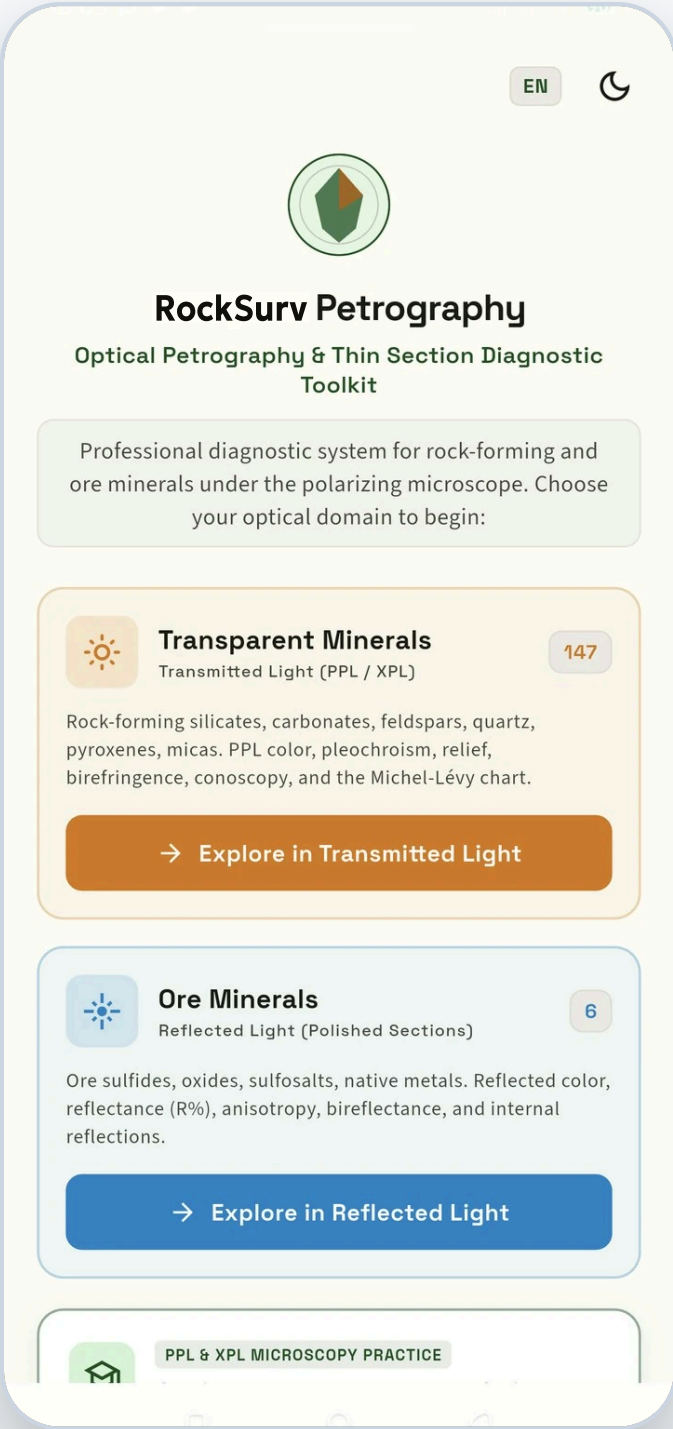
Persistent header toggle seamlessly reconfigures diagnostic keys, search catalogs, reference tables, and micrograph galleries.

- PPL / XPL (147 Minerals)

● RLM / XRL (Ore Phases)

● Instant Mode Toggle

PPL/XPL and RLM/XRL workflows are unified in a single responsive interface calibrated for standard polarizing microscopes.



10-Parameter Optical Key for Thin Section Identification

Instantaneous multi-parameter filtering under the microscope: each entered observation narrows the candidate list in real time.

Microscopic Observation Suite

Crystal system, relief and Becke line, color, pleochroism, cleavage, extinction angle, birefringence, elongation, optic sign, and 2V angle.

Ergonomic Chip Selectors

Fast one-tap chips designed for rapid input during active microscope stage manipulation and simultaneous micrometer focusing.

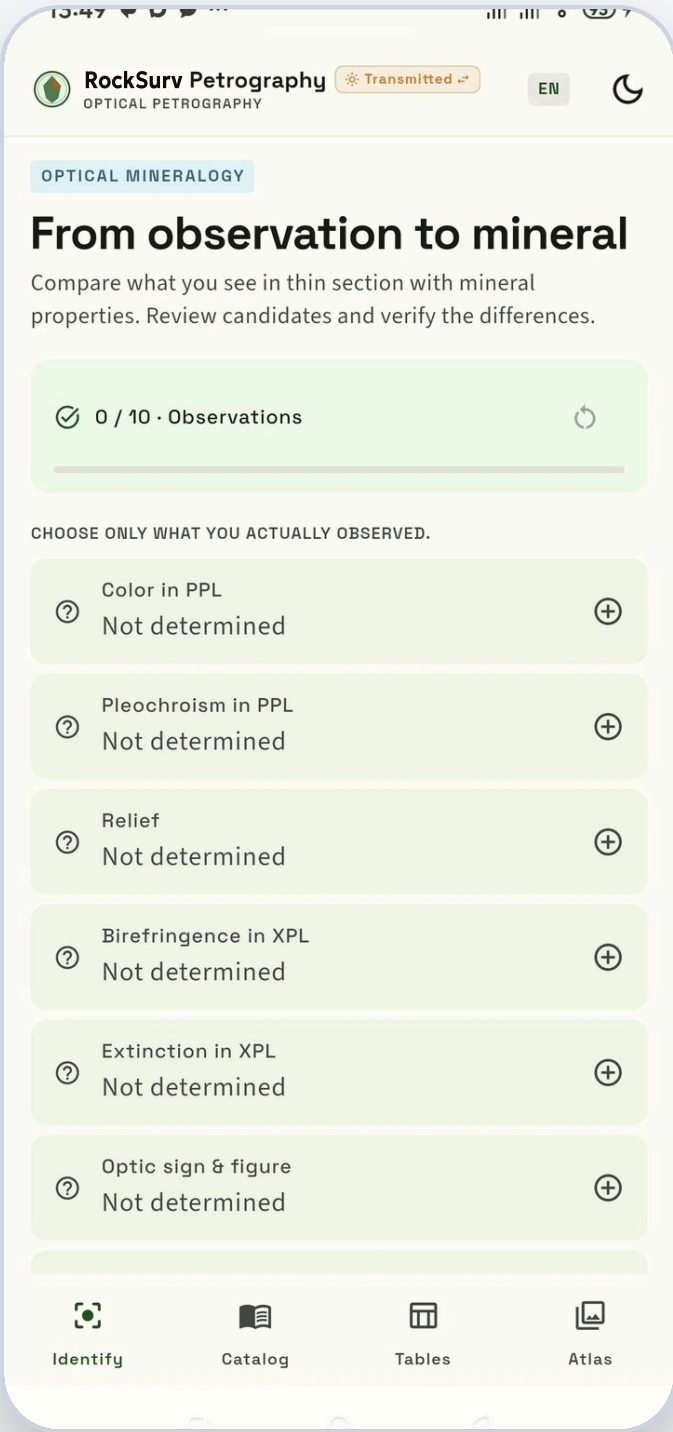
Dynamic Candidate Counter

Live candidate counter immediately reflects remaining matches, guiding the petrographer toward the next decisive diagnostic test.

- 10 Optical Criteria

• Real-Time Filtering

• Stage-Side Ergonomics



The diagnostic engine matches observations at the microscope stage against certified optical databases with sub-millisecond response.

Explainable Deterministic Matching with Candidate Ranking

Transparent mathematical scoring ranks candidates by likelihood, distinguishing verified matches from neutral criteria and direct contradictions.

Transparent Score Calculation

Verified matches add positive score (+1), unobserved properties remain neutral (0), and incompatible traits apply penalties (-1).

Confidence Percentage

Clear percentage ranking displays proximity to ideal mineral match (Prehnite 82%, Augite 67%, Olivine 58%), robust against incomplete datasets.

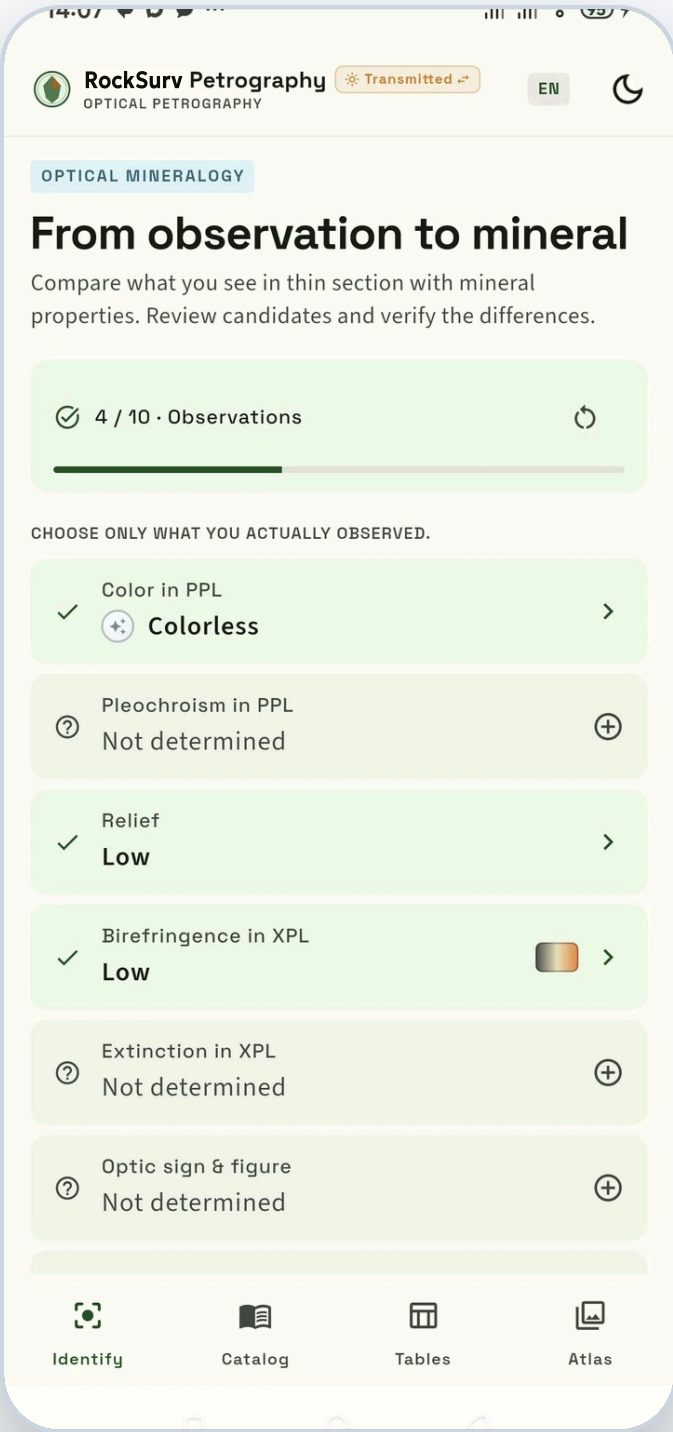
Detailed Diagnostic Explanation Sheet

Expandable breakdown reveals every optical parameter, showing precisely why each mineral was matched or excluded.

- Deterministic Scoring

• Ranked by %

• Detailed Explanation Sheet



No 'black box' guessing: petrographers inspect the exact mathematical contribution of every observed optical property.

Curated Thin Section Atlas: 267+ High-Resolution Micrographs

Comprehensive reference gallery of authentic thin sections with verified geological provenance and dual-illumination metadata.

Lighting Mode Filter Chips

Instant filtering by All, PPL (Plane-Polarized Light), XPL (Cross-Polarized Light), and Paired synchronized micrographs.

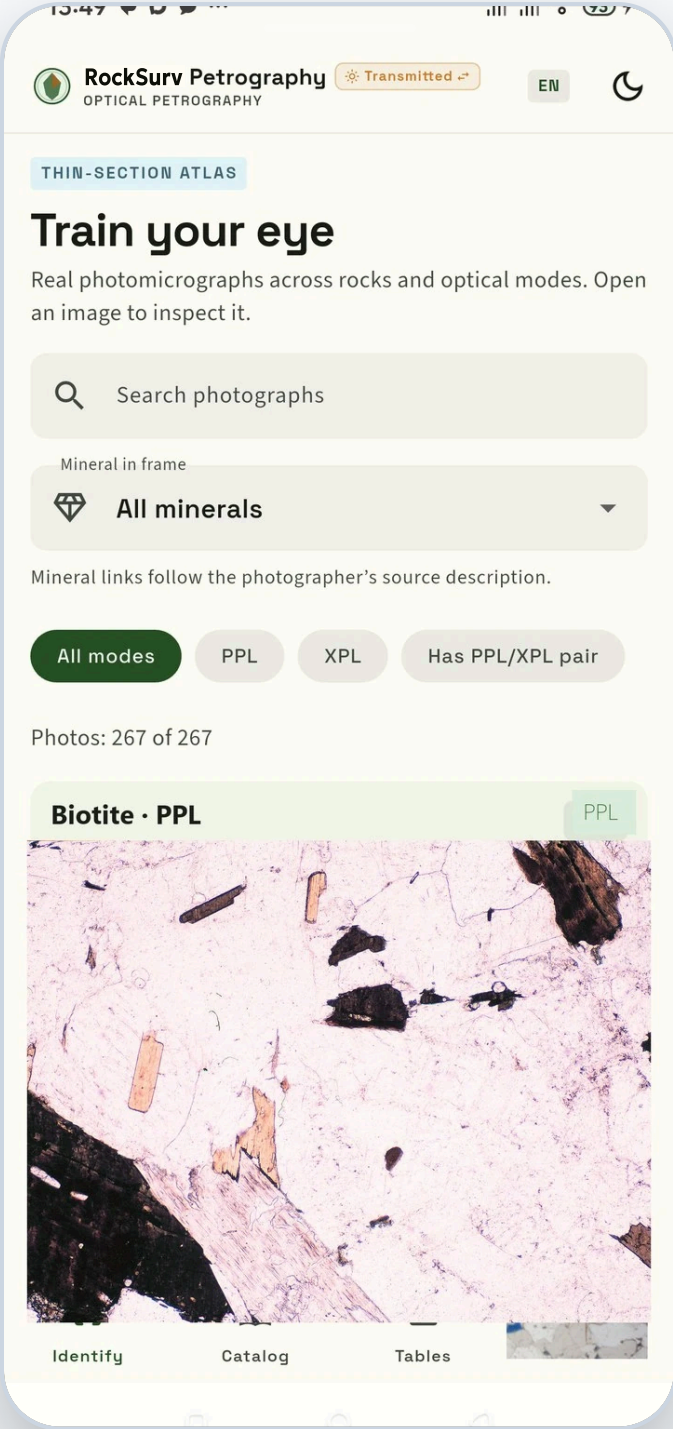
Calibrated 1 mm Optical Scale Bars

Embedded metric scale bars ensure accurate assessment of grain size, mineral intergrowths, and micro-textural fabrics.

Petrographic Associations

Clear mineral labeling and contextual rock descriptions across igneous, metamorphic, and sedimentary host lithologies.

- 267+ Thin Section Shots
- PPL / XPL / Pairs Filters
- 1 mm Scale Bar



Every micrograph in the atlas includes certified provenance, lithological classification, and mineral paragenesis notes.

Synchronized PPL & XPL Analysis: Pleochroism & Birefringence

Side-by-side comparison of the same grain field under plane- and crossed-polarized light reveals twin laws, alteration, and zoning.

High-Resolution Zoom Viewer

Smooth pan and zoom photomicrograph inspection with calibrated 1 mm metric scale bar for granular fabric analysis.

Diagnostic Contrast

Direct comparison of natural mineral color/pleochroism in PPL with interference colors, twinning, and extinction in XPL.

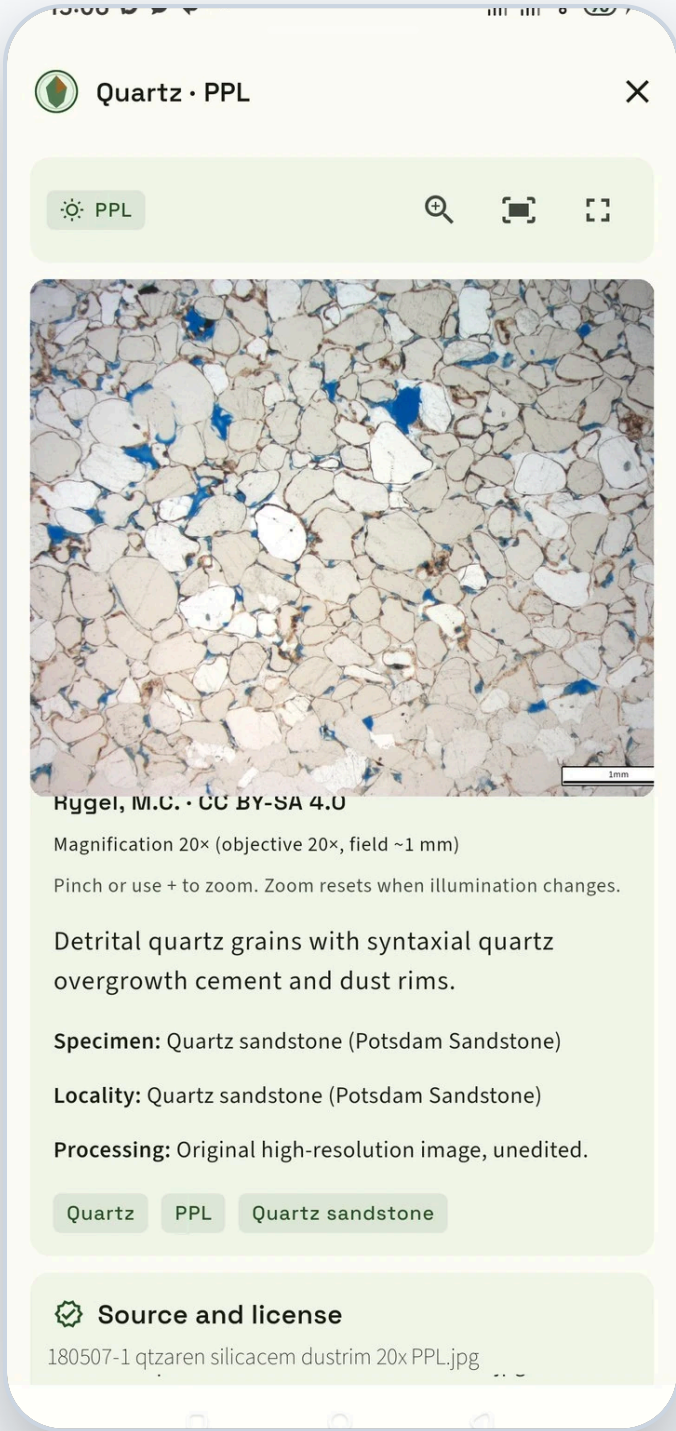
Academic Citation & Sample Provenance

Complete thin section metadata, locality, petrographic context, and sample source accessible in one click.

• Synchronized PPL/XPL Pairs

• Interference & Twinning

• 1 mm Metric Scale Bar



Paired comparison reliably exposes polysynthetic plagioclase twinning, microcline cross-hatching, and exsolution lamellae.

Interactive Michel-Lévy Nomogram & Birefringence Solver

Real-time optical solver connecting standard thin section thickness (30 μm), optical retardation, and birefringence (Δn).

Formula-Driven Precision

Computes retardation $\Gamma = 1000 \cdot d \cdot \Delta n$ with an interactive slider across 1st to 4th order interference color spectra.

Standard Slide Thickness

Defaults to standard 30 μm slide thickness with manual micrometer override for wedge-shaped or non-standard cuts.

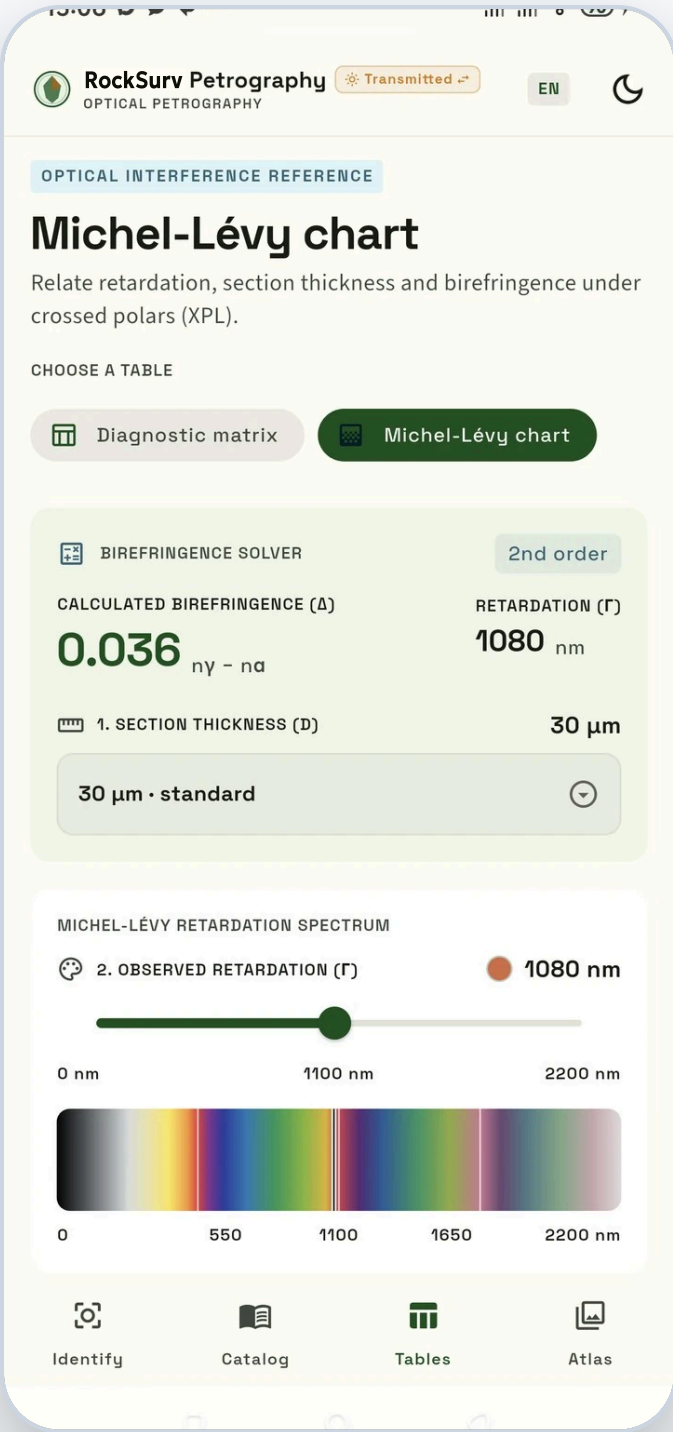
Candidate Mineral Filtering

Instantaneous list of rock-forming minerals whose maximum birefringence matches the selected interference color band.

- Formula $\Gamma = 1000 \cdot d \cdot \Delta n$

• 30 μm Standard Cut

• Mineral Auto-Match



Precise digital counterpart of the classic Michel-Lévy nomogram for rapid birefringence calculation at the microscope.

Deep Mineral Profiles: Crystallography & Optical Constants

Exhaustive mineralogical dossiers covering refractive indices ($n\alpha$, $n\beta$, $n\gamma$), 2V angle, indicatrix geometry, and paragenesis.

Complete Optical Dossier

Birefringence Δn , optical sign (+/-), optic axial angle 2V, dispersion, pleochroic scheme, and extinction characteristics.

Crystallography & Chemical Formula

Crystal system, space group, ideal chemical formula, Mohs hardness, and cleavage plane orientations.

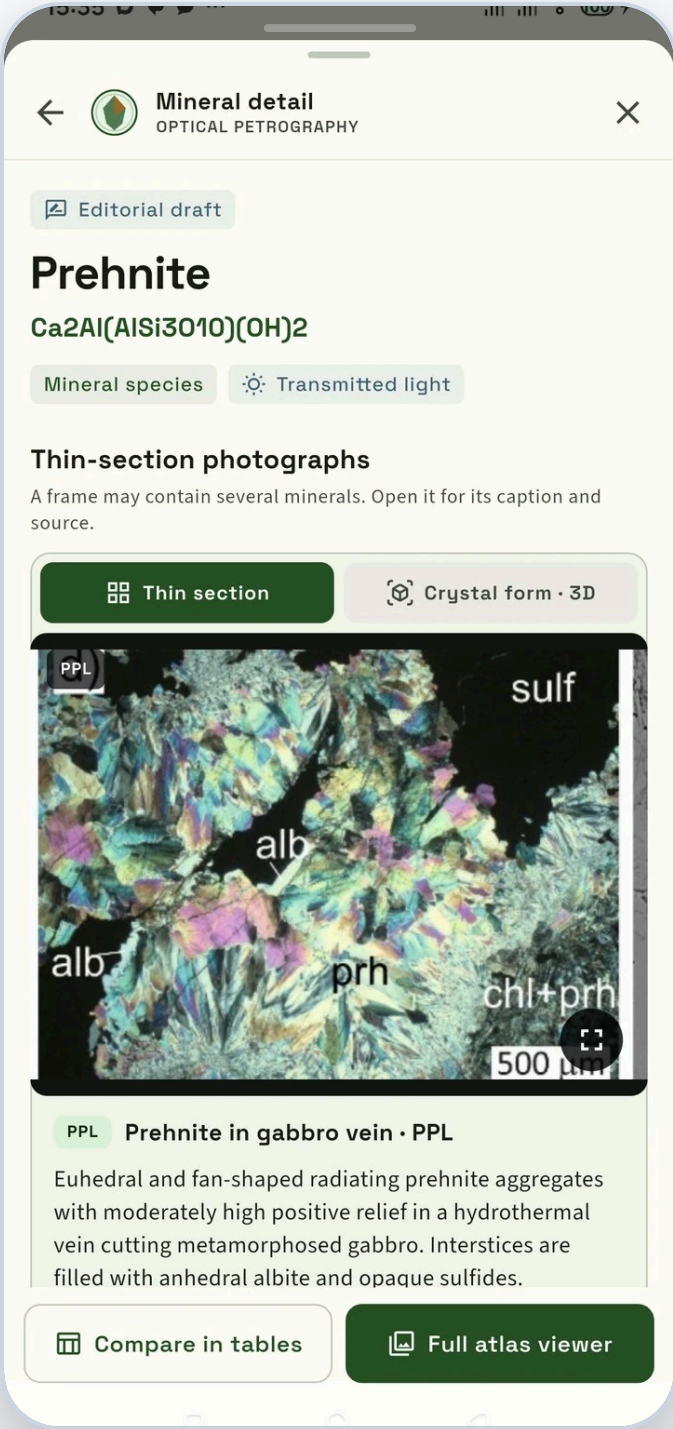
Petrogenetic Associations

Diagnostic crystallization conditions, mineral co-occurrences, and characteristic alteration products (sericite, chlorite, saussurite).

- Optical Indicatrix

• Refractive Indices

• Paragenesis & Symmetry



Every profile integrates verified optical constants, structural formulas, and diagnostic differentiators from common lookalikes.

Reflected Light Microscopy: Opaque & Ore Mineral Diagnostics

Dedicated module for opaque ore minerals in polished sections: reflectance (R%), bireflectance, anisotropy, and internal reflections.

Reflectance & Color in Air and Oil

Quantitative R% values and distinctive tints for sulfides and oxides (pyrite, chalcopyrite, galena, sphalerite).

Anisotropy & Internal Reflections

Evaluation under crossed polars (XRL) for reliable differentiation of isotropic and anisotropic ore mineral phases.

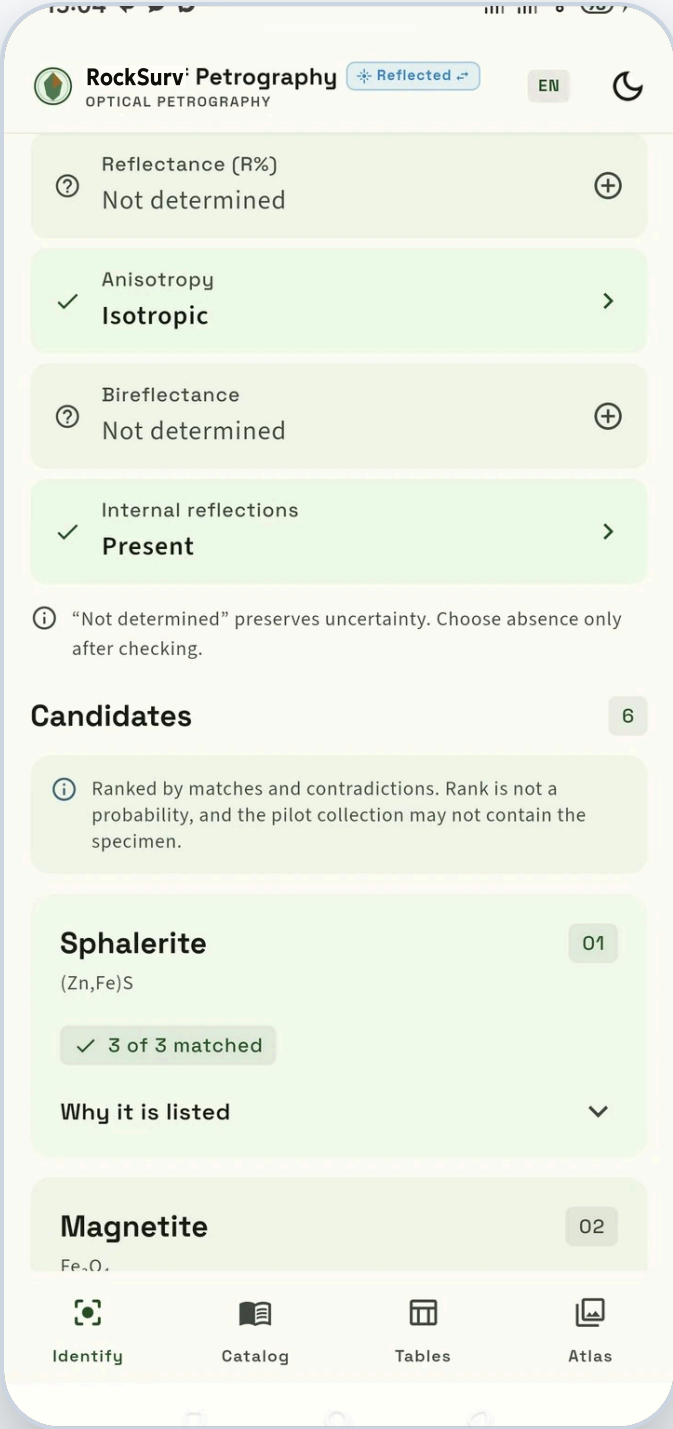
Polishing Hardness & Textural Intergrowths

Relative polishing relief, grain morphologies, and diagnostic solid-solution exsolution textures in polished mounts.

- Reflectance R%

• Sulfides & Oxides

• Anisotropy in XRL



Ore mineral identification follows standardized ore microscopy protocols for polished thin sections and core mounts.

Optical Practice Lab: Interactive Microscopic Simulators

Hands-on interactive training tools simulating the Becke line test, relief changes, pleochroism rotation, and extinction angles.

Becke Line & Relief Simulator

Interactive focus defocusing demonstrating Becke line migration into the medium of higher refractive index.

Rotating Stage & Extinction Angle Simulator

Full 360° stage rotation simulating parallel, inclined, and undulose extinction under crossed polarized light (XPL).

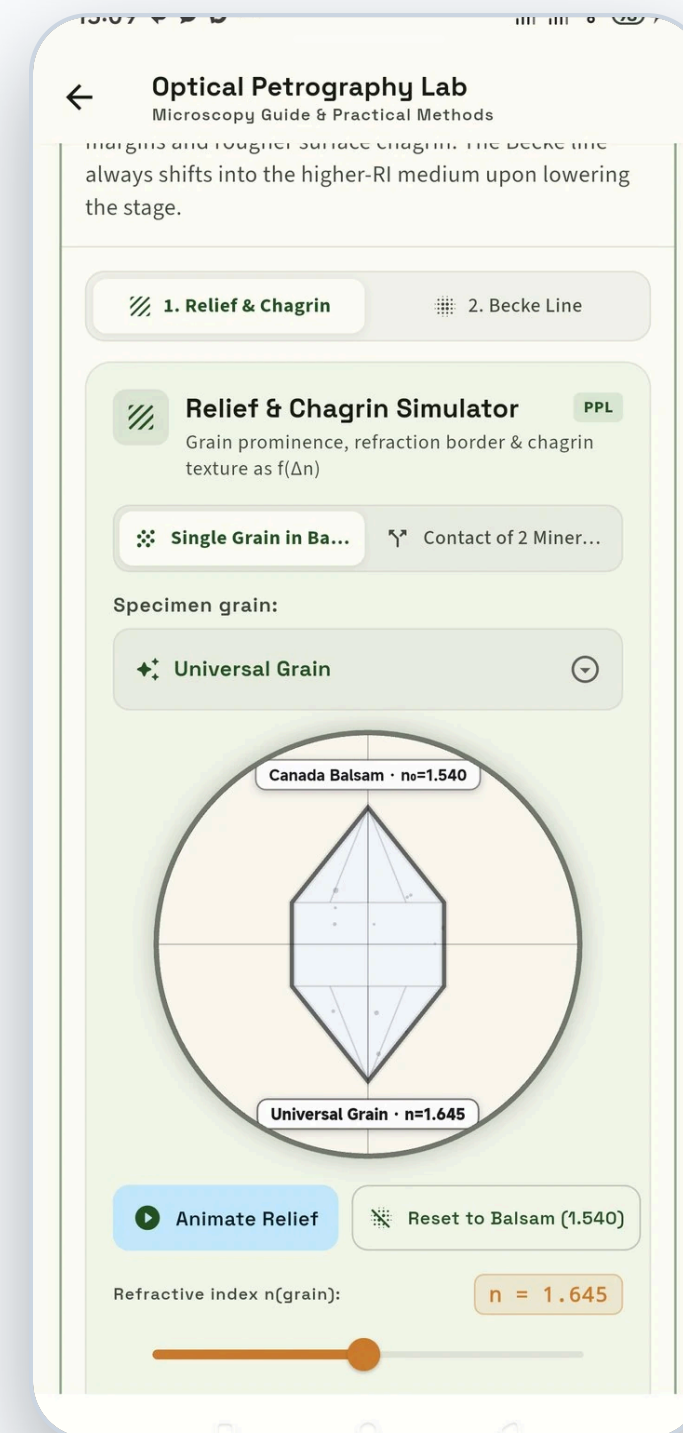
Pleochroism & Absorption Scheme Simulator

Full 360° stage rotation demonstrating anisotropic absorption color shift for direct (biotite $N_g > N_p$) and reverse (tourmaline) schemes.

• Becke Line & Relief

• 360° Stage Rotation

• Pleochroism & Absorption



The virtual laboratory bridges the gap between petrographic theory and real polarizing microscope operations.

Multilingual Architecture & High-Contrast Petrographic UX

Localized in 4 languages including full native RTL layout for Arabic, paired with a darkfield visual theme optimized for microscopy lab lighting.

4 Core Languages

Complete user interface localization in English, Russian, Arabic (with mirrored RTL layout), and Chinese.

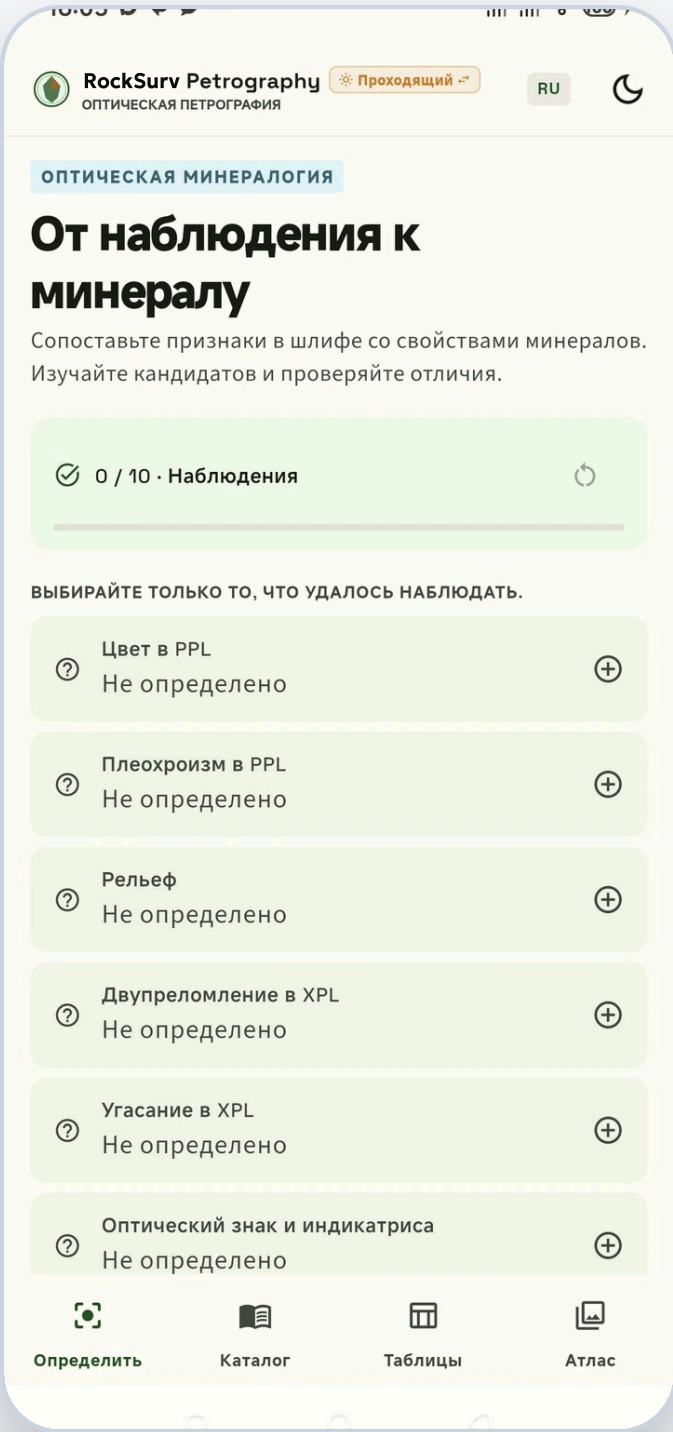
Darkfield Optical UI Theme

Engineered to minimize retinal fatigue during extended observation sessions in darkened petrographic laboratories.

Touch Targets & Ergonomics

All touch targets meet or exceed 48 dp for reliable, error-free interaction beside the microscope stage.

- 4 Languages (EN, RU, AR RTL, ZH)
 - Darkfield Optical Theme
 - Touch Targets >= 48dp



Engineered for international geological research teams and adapted to low-light laboratory operating environments.

100% Offline Autonomy: Engineered for Field Camp & Lab

Completely self-contained databases and high-resolution assets: zero cloud dependency, zero latency, and mission-critical reliability.

Zero-Latency Local Engine

All 147+ transparent and 6 ore minerals, 267+ micrographs, and interactive simulators operate entirely on-device.

Remote Field Readiness

Operates without degradation in remote exploration camps, core sheds, and fly-in geological field expeditions with no cellular coverage.

Academic & Production Standard

Designed for field geologists, petrographers, mineralogists, university researchers, and exploration mining teams.

- 100% Standalone Offline

• Zero Latency Engine

• Academic Standard



RockSurv Petrography delivers the comprehensive power of an atlas, deterministic key, and optical lab in a single standalone app.